

Mathematics for Computer Science – 2

Libin Wang

School of Computer Science, South China Normal University

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Definition of congruence (同余) .

“同余” 这个概念是 CSers 的好朋友。

Definition

We say that a is congruent to b modulo m , and we write

$$a \equiv b \pmod{m},$$

if m divides $a - b$. The number m is called the *modulus* (模数) of the congruence.

Some notations.

$$a \equiv b \pmod{m} \iff$$

- $\exists k \in \mathbb{Z}, a = km + b.$
- $(a \bmod m) = (b \bmod m)$
- $m \mid (a - b)$

Some notations.

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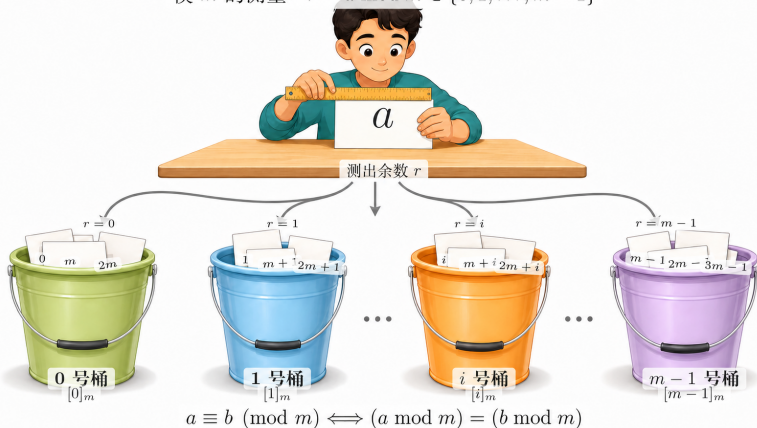
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提示.

表达是思维的指针。

同余的直观

模 m 的测量: $r = a \bmod m \in \{0, 1, \dots, m-1\}$



Example

$$26 \equiv 8 \pmod{9} \quad \text{and} \quad 6 \equiv 55 \pmod{7},$$

since

$$9 \mid (26 - 8) \quad \text{and} \quad 7 \mid (6 - 55),$$

or, equivalently:

$$8 = 26 - 2 \cdot 9 \quad \text{and} \quad 55 = 6 + 7 \cdot 7.$$

Lemma

If $a_1 \equiv b_1 \pmod{m}$ and $a_2 \equiv b_2 \pmod{m}$, then

$$a_1 \pm a_2 \equiv b_1 \pm b_2 \pmod{m}$$

and

$$a_1 a_2 \equiv b_1 b_2 \pmod{m}$$

课堂练习!

练习

假设 $x \equiv a \pmod{n}$, 且 $n = p \cdot q$, p, q 是两个素数。请证明, 因此必然有 $x \equiv a \pmod{p}$ 和 $x \equiv a \pmod{q}$ 。

Examples

Since $10000 \equiv 1 \pmod{3}$ and $998 \equiv 2 \pmod{3}$, then

$$10000 \cdot 998 \equiv 2 \pmod{3}.$$

练习

计算以下等式：

$$7^{500} \bmod 8$$

$$1! + 2! + 3! + \cdots + 99! + 100! \bmod 12$$

$$2^{20} - 1 \bmod 41$$

练习

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答案

1、9、0。

Properties of Congruence.

Negative number.

Let x and n be two positive integers and $x < n$, what does $-x \bmod n$ mean?

Properties of Congruence.

Negative number.

Let x and n be two positive integers and $x < n$, what does $-x \bmod n$ mean?

Intuition.

Consider that the negative of x is the number x' such that $x + x' = 0$, that is:

$$x + x' \equiv 0 \pmod{n}.$$

Since $x < n$, hence $x' = n - x$.

Two's Complement(二进制补码)

Two's Complement

A signed number represented in n bits. The range of the numbers is $[-2^{n-1}, 2^{n-1} - 1]$, and the rule is described as follows:

- Positive integers, in the range 0 to $2^{n-1} - 1$, are stored in regular binary form. The sign bit is set to 0.
- Negative integers $-x$, with $1 \leq x \leq 2^{n-1}$, are calculated by first constructing x in binary, then inverting all the bits of x and finally adding 1. The sign bit is set to 1.

Two's Complement

Another way to remember two's complement.

The negative of x equals $2^n - x$. Since $2^n - x = \underbrace{111 \cdots 11}_n + 1 - x$,
and $\underbrace{111 \cdots 11}_n - x$ is the same as inverting all the bits of x .

Theorem (Cancellation Law.)

If $\gcd(c, m) = 1$ and

$$ac \equiv bc \pmod{m},$$

then

$$a \equiv b \pmod{m},$$

where gcd shorts for the greatest common divisor.

Cancellation Law.

Proof.

By definition of congruence, we have $m|(ac - bc)$, equivalently, $m|(a - b)c$. Since $\gcd(c, m) = 1$, it follows that $m \mid (a - b)$, so as claimed.



Another perspective.

If $\gcd(c, m) = 1$ then $\exists r, s \in \mathbb{Z}$ s.t.

$$rc + sm = 1$$

both sides of the equation modulo m , we have:

$$rc \equiv 1 \pmod{m}$$

means r is the multiplicative inverse(乘法逆元) of $c \bmod m$, let it be c^{-1} .

Partially solve the congruence $ax \equiv b \pmod{m}$.

Example

To solve $3x \equiv 2 \pmod{11}$.

Firstly, by using egcd algorithm, to compute that $3^{-1} = 4$, because $3 \cdot 4 \equiv 1 \pmod{11}$. Multiply 4 to the equation and obtain

$$x \equiv 8 \pmod{11}.$$

问题.

消去律源自于同余的定义与欧几里得引理。可以执行消去操作是因为存在唯一乘法逆元，乘法逆元源自于 Bezout 定理，但是 Bezout 系数不唯一。请问，为什么乘法逆元会唯一呢？

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回答.

需要重新审视 Bezout 系数的形式。

Properties of Congruence.

Lemma

For $n \in \mathbb{N}$, congruence modulo n forms an equivalence relation(等价关系) of $\mathbb{Z}$.

Proof.

It is easy to check that:

1. Reflexive(自反性). $a \equiv a \pmod{n}$
2. Symmetric(对称性). If $a \equiv b \pmod{n}$ then $b \equiv a \pmod{n}$
3. Transitive(传递性). If $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$ then $a \equiv c \pmod{n}$ □

问题

等价关系意味着什么？

问题

等价关系意味着什么？

条件反射式回答.

存在等价类，即对数据的划分 (Partition)。

Equivalence relation and equivalence classes.

Definition

When a set S has an equivalence relation on it, then the equivalence relation partitions (划分、分割) the set S into disjoint subsets, called equivalence classes (等价类), defined by the property that two elements are in the same equivalence class if they are equivalent.

Congruence classes modulo m .

The set of congruence classes modulo m is denoted by $\mathbb{Z}/m\mathbb{Z}$.
There are exactly m congruence classes in $\mathbb{Z}/m\mathbb{Z}$. That is :

$$\mathbb{Z}/m\mathbb{Z} = \{[0]_m, [1]_m, \dots, [m-1]_m\}.$$

Example

When $m = 2$, $\mathbb{Z}/2\mathbb{Z} = \{[0]_2, [1]_2\}$. The congruence class $[1]_2$ is the set of all integers congruent to 1 modulo 2. Thus $[1]_2$ is the set of all odd integers. Similarly, the congruence class $[0]_2$ is the set of all even integers.

Proposition of Congruence.

Proposition

If $[a_1]_m = [a_2]_m$ and $[b_1]_m = [b_2]_m$, then

$$[a_1 \pm b_1]_m = [a_2 \pm b_2]_m, \quad \text{and} \quad [a_1 b_1]_m = [a_2 b_2]_m.$$

Proof.

It is easy. Transform the form $[a]_m = [b]_m$ to $a \equiv b \pmod{m}$ and use Proposition 2.2. □

Notations of Congruence classes modulo m .

- Any element b of a congruence class $[a]_m$ is called a *representative* (代表元) of that class.
- The set of all the least nonnegative representative of $\mathbb{Z}/m\mathbb{Z}$ is the set of integers $\{0, 1, 2, \dots, m-1\}$, that is called the *least residue system* modulo m (模 m 的最小剩余系).
- Any set of m integers, no two of which are congruent modulo m , is called a *complete residue system* modulo m (模 m 的完全剩余系).

Notations of Congruence classes modulo m .

Example

Let $m = 7$, the least residue system modulo m is the set $\{0, 1, 2, 3, 4, 5, 6\}$, and a complete residue system modulo m may be the set $\{14, 8, 23, 46, 61, 13\}$.

In this section, we focus on modular exponentiation which is an important arithmetic primitive. Its task is that given integers x , y and m to compute

$$x^y \bmod m.$$

Example

To compute $2^{16} \bmod 11$. We compute:

$$2^2 \bmod 11 = 4$$

$$2^4 \bmod 11 = 4 \cdot 4 \bmod 11 = 5$$

$$2^8 \bmod 11 = 5 \cdot 5 \bmod 11 = 3$$

$$2^{16} \bmod 11 = 3 \cdot 3 \bmod 11 = 9$$

Example

To compute $2^{22} \bmod 11$. We compute:

$$2^2 \bmod 11 = 4$$

$$2^4 \bmod 11 = 4 \cdot 4 \bmod 11 = 5$$

$$2^8 \bmod 11 = 5 \cdot 5 \bmod 11 = 3$$

$$2^{16} \bmod 11 = 3 \cdot 3 \bmod 11 = 9$$

then

$$(2^{22} = 2^2 \cdot 2^4 \cdot 2^{16}) \equiv 4 \cdot 5 \cdot 9 \equiv 4 \pmod{11}$$

The process can be expressed as a recursive form, by that, we sharply improve the efficiency from performing $O(y)$ multiplications to $O(\log(y))$.

$$x^y = \begin{cases} (x^{\lfloor y/2 \rfloor})^2 & \text{if } y \text{ is even;} \\ x \cdot (x^{\lfloor y/2 \rfloor})^2 & \text{if } y \text{ is odd.} \end{cases} \quad (1)$$

Listing 1: Recursive Modular Exponentiation

```
1  # Recursive Function to calculate  
2  #  $(x^y) \% p$  in  $O(\log y)$   
3  def rec_mod_exp(x, y, p):  
4      if (y == 0): return 1  
5      z = rec_mod_exp(x, y//2, p)  
6      if ((y & 1) == 0): #y is an even number  
7          return z*z % p  
8      else: #y is an odd number  
9          return x*z*z % p
```

Mod Exponentiation: from recursive to iterative.

We describe how to transform the recursive algorithm to an iterative algorithm as follows. Firstly, we treat integer y as a polynomial (or a binary string):

$$y = y_{n-1}2^{n-1} + y_{n-2}2^{n-2} \cdots + y_12 + y_0,$$

where $y_i \in \{0, 1\}$.

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Secondly, transform x^y as:

$$x^y = \prod_{i=0}^{n-1} x^{y_i 2^i}$$

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Secondly, transform x^y as:

$$x^y = \prod_{i=0}^{n-1} x^{y_i 2^i}$$

Finally, start with x and repeatedly square modulo m , multiply the terms with $y_i = 1$ and get the result.

Mod Exponentiation: Example.

A concrete example.

Let $x = 7$, $y = 10$, $m = 11$, to compute $x^y \bmod m$. The binary string of y is 1010, thus we compute:

$$y_0 = 0, \quad x^{2^0} \equiv 7 \pmod{m}$$

$$y_1 = 1, \quad x^{2^1} \equiv 5 \pmod{m}$$

$$y_2 = 0, \quad x^{2^2} \equiv 3 \pmod{m}$$

$$y_3 = 1, \quad x^{2^3} \equiv 9 \pmod{m}$$

Then, multiply the terms with $y_i = 1$, we have

$$x^y = (5 \cdot 9) \bmod 11 = 1$$

练习.

Let $x = 5$, $y = 13$, $m = 13$, to compute $x^y \bmod m$.

练习.

Let $x = 5$, $y = 13$, $m = 13$, to compute $x^y \bmod m$.

答案.

5.

Mod Exponentiation (Iterative version).

Because the white board is too narrow to show the code, so it is your home work.

Some topics using modular arithmetic.

Our following job is to play with number using modular arithmetic, and find some patterns or rules.

Find the patterns.

Let $p = 7$, and for every $1 \leq a < p$, compute $a^i \bmod p$, where $1 \leq i < p$. We have:

a	a^2	a^3	a^4	a^5	a^6
1	1	1	1	1	1
2	4	1	2	4	1
3	2	6	4	5	1
4	2	1	4	2	1
5	4	6	2	3	1
6	1	6	1	6	1

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2	4	1	2	4	1
3	2	6	4	5	1
4	2	1	4	2	1
5	4	6	2	3	1
6	1	6	1	6	1

May you find some patterns?

More data to find the patterns.

Let $p = 11$, and for every $1 \leq a < p$, compute $a^i \bmod p$, where $1 \leq i < p$. We have:

a	a^2	a^3	a^4	a^5	a^6	a^7	a^8	a^9	a^{10}
1	1	1	1	1	1	1	1	1	1
2	4	8	5	10	9	7	3	6	1
3	9	5	4	1	3	9	5	4	1
4	5	9	3	1	4	5	9	3	1
5	3	4	9	1	5	3	4	9	1
6	3	7	9	10	5	8	4	2	1
7	5	2	3	10	4	6	9	8	1
8	9	6	4	10	3	2	5	7	1
9	4	3	5	1	9	4	3	5	1
10	1	10	1	10	1	10	1	10	1

猜想

For p is a prime, and $\gcd(a, p) = 1$,

$$a^{p-1} \equiv 1 \pmod{p}$$

Another computation.

$\forall 1 < a < p$, compute $a \cdot i \bmod p$, for $1 \leq i < p$. For example, let $a = 2$, $p = 7$, we have:

$a \cdot i$	1	2	3	4	5	6
$a = 1$	1	2	3	4	5	6
$a = 2$	2	4	6	1	3	5

Another computation.

Continue the computation...

$a \cdot i$	1	2	3	4	5	6
$a = 1$	1	2	3	4	5	6
$a = 2$	2	4	6	1	3	5
$a = 3$	3	6	2	5	1	4

Another computation.

Continue the computation...

$a \cdot i$	1	2	3	4	5	6
$a = 1$	1	2	3	4	5	6
$a = 2$	2	4	6	1	3	5
$a = 3$	3	6	2	5	1	4

Have a decision?

Another computation.

Finally!

$a \cdot i$	1	2	3	4	5	6
$a = 1$	1	2	3	4	5	6
$a = 2$	2	4	6	1	3	5
$a = 3$	3	6	2	5	1	4
$a = 4$	4	1	5	2	6	3
$a = 5$	5	3	1	6	4	2
$a = 6$	6	5	4	3	2	1

Conjecture.

The trick is, p is a prime number! We conjecture: if p is a prime, $\forall a$ which is not divided by p , $a \cdot i \bmod p$, for $1 \leq i < p$, is a permutation of numbers from 1 to $p - 1$.

Conjecture.

The trick is, p is a prime number! We conjecture: if p is a prime, $\forall a$ which is not divided by p , $a \cdot i \bmod p$, for $1 \leq i < p$, is a permutation of numbers from 1 to $p - 1$. That is, the numbers

$$a, 2a, 3a, \dots, (p-1)a \pmod{p}$$

are the same as the numbers:

$$1, 2, 3, \dots, p-1$$

although they may be in a different order.

Conjecture.

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although they may be in a different order.

$$S = \{a \cdot i \bmod p, 1 \leq i < p\}$$

is also called a complete system of residues modulo p .

Proof by contradiction.

Of course, we need a proof!

Proof by contradiction.

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Proof.

Proof by contradiction (informal and incomplete). If we are wrong, then there exist i and j such that,

$$a \cdot i \equiv a \cdot j \pmod{p}$$

where $i \neq j$. However, then we can cancel the a from the equation! (Cancellation Law.) □

Do a simple job!

Multiply all $1 \leq i < p$, and all $a \cdot i \bmod p$, we have:

$$\prod_{i=1}^{p-1} i = \prod_{i=1}^{p-1} (a \cdot i \bmod p)$$

Do a simple job!

Multiply all $1 \leq i < p$, and all $a \cdot i \bmod p$, we have:

$$\prod_{i=1}^{p-1} i = \prod_{i=1}^{p-1} (a \cdot i \bmod p)$$

Convince yourself:

$$\prod_{i=1}^{p-1} i \equiv \prod_{i=1}^{p-1} a \cdot i \equiv a^{p-1} \prod_{i=1}^{p-1} i \pmod{p}$$

Do a simple job!

Multiply all $1 \leq i < p$, and all $a \cdot i \bmod p$, we have:

$$\prod_{i=1}^{p-1} i = \prod_{i=1}^{p-1} (a \cdot i \bmod p)$$

Convince yourself:

$$\prod_{i=1}^{p-1} i \equiv \prod_{i=1}^{p-1} a \cdot i \equiv a^{p-1} \prod_{i=1}^{p-1} i \pmod{p}$$

Cancel the big number, we have:

$$a^{p-1} \equiv 1 \pmod{p}$$

Fermat's little theorem.

Theorem (Fermat's little theorem.)

Let p be a prime number, and let a be any number with $a \not\equiv 0 \pmod{p}$. Then

$$a^{p-1} \equiv 1 \pmod{p}.$$

An Exercise using Fermat's little theorem.

Exercise.

Let $p = 17$ be a prime number, and let $a = 2$, what is $a^{2024} \bmod p$?

An Exercise using Fermat's little theorem.

Exercise.

Let $p = 17$ be a prime number, and let $a = 2$, what is $a^{2024} \bmod p$?

$$2^{2024} \equiv 2^{126 \cdot 16 + 8} \equiv 2^8 \equiv 1 \pmod{17}.$$

Exercises using Fermat's little theorem.

Exercises.

- $a \equiv 9^{794} \pmod{73}$, find a and $0 \leq a \leq 73$.
- Solve $x^{87} \equiv 6 \pmod{29}$.

More computation.

If the modulus is a composite number, then our trick will fail! For example, let $n = 6$,

$a \cdot i$	1	2	3	4	5
$a = 1$	1	2	3	4	5
$a = 2$	2	4	0	2	4
$a = 3$	3	0	3	0	3
$a = 4$	4	2	0	4	2
$a = 5$	5	4	3	2	1

One more computation.

Let $n = 9$,

$a \cdot i$	1	2	3	4	5	6	7	8
$a = 1$	1	2	3	4	5	6	7	8
$a = 2$	2	4	6	8	1	3	5	7
$a = 3$	3	6	0	3	6	0	3	6
$a = 4$	4	8	3	7	2	6	1	5
$a = 5$	5	1	6	2	7	3	8	4
$a = 6$	6	3	0	6	3	0	6	3
$a = 7$	7	5	3	1	8	6	4	2
$a = 8$	8	7	6	5	4	3	2	1

观察.

If n is a composite number, the numbers

$$a, 2a, 3a, \dots, (n-1)a \pmod{n}$$

may **NOT** be the same as the numbers:

$$1, 2, 3, \dots, n-1$$

Check the observation.

Let $n = 9$, for all $i \in [1, n - 1]$, and $\gcd(i, n) = 1$:

$a \cdot i$	1	2	4	5	7	8
$a = 1$	1	2	4	5	7	8
$a = 2$	2	4	8	1	5	7
$a = 4$	4	8	7	2	1	5
$a = 5$	5	1	2	7	8	4
$a = 7$	7	5	1	8	4	2
$a = 8$	8	7	5	4	2	1

Conjecture.

We conjecture: Let n be a composite number, denotes

$$S = \{b : 1 \leq b < n \text{ and } \gcd(b, n) = 1\}$$

Then $\forall a$ with $\gcd(a, n) = 1$, denotes

$$S' = a \cdot S \pmod{n}$$

we have:

$$S = S'$$

Euler's phi function.

Define:

$$\varphi(n) = |\{b : 1 \leq b < n \text{ and } \gcd(b, n) = 1\}|$$

The function φ is called *Euler's phi function*.

Then:

$$S = \{b_1, b_2, \dots, b_{\varphi(n)} : 1 \leq b_i < n \text{ and } \gcd(b_i, n) = 1\}$$

$$S' = \{a \cdot b_1, a \cdot b_2, \dots, a \cdot b_{\varphi(n)} \pmod{n} : \\ b_i \in S \text{ and } \gcd(a, n) = 1\}.$$

To prove

$$S = S'$$

Check that if there exist:

$$a \cdot b_i \equiv a \cdot b_j \pmod{n}$$

where $b_i \neq b_j$. Then by Cancellation Law,

$$b_i \equiv b_j \pmod{n}.$$

Contradiction!

Do a similar simple job!

Multiply all the numbers in S and S' , we have:

$$\prod_{i=1}^{\varphi(n)} b_i = \prod_{i=1}^{\varphi(n)} (a \cdot b_i \bmod n)$$

$$\prod_{i=1}^{\varphi(n)} b_i \equiv \prod_{i=1}^{\varphi(n)} a \cdot b_i \pmod{n}$$

$$\prod_{i=1}^{\varphi(n)} b_i \equiv a^{\varphi(n)} \prod_{i=1}^{\varphi(n)} b_i \pmod{n}$$

Cancel the big number, we have:

$$a^{\varphi(n)} \equiv 1 \pmod{n}$$

Euler's Theorem.

(Euler's Theorem.)

Let n be a positive composite number, a be a positive integer with $\gcd(a, n) = 1$, then:

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

例.

设 $n = 100$, 则 $\varphi(n) = 40$, 所以 $7^{40} \equiv 1 \pmod{n}$ 。同样, $9^{40} \equiv 1 \pmod{n}$ 。

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问题.

为什么 $\varphi(n) = 40$? 或者, 给定 n , 如何求 $\varphi(n)$?

More about Euler's Phi Function.

Definition (Euler's Phi Function.)

Define:

$$\varphi(n) = |\{b : 1 \leq b < n \text{ and } \gcd(b, n) = 1\}|$$

The function φ is called *Euler's phi function*.

Observations

$\varphi(p) = p - 1$, where p is a prime.

$\varphi(p^k) = p^k - p^{k-1}$, where p is a prime.

More about Euler's Phi Function..

Question.

How to compute $\varphi(m)$ where $m = p^i q^j$ with p and q are prime.

More about Euler's Phi Function..

Question.

How to compute $\varphi(m)$ where $m = p^i q^j$ with p and q are prime.

A related question.

How to compute $\varphi(mn)$ where $\gcd(m, n) = 1$.

More about Euler's Phi Function.

Compute $\varphi(mn)$

Display the positive integers not exceeding mn in the following way.

1	$m + 1$	$2m + 1$	$\dots$	$(n - 1)m + 1$
2	$m + 2$	$2m + 2$	$\dots$	$(n - 1)m + 2$
3	$m + 3$	$2m + 3$	$\dots$	$(n - 1)m + 3$
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$
r	$m + r$	$2m + r$	$\dots$	$(n - 1)m + r$
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$
m	$2m$	$3m$	$\dots$	mn

More about Euler's Phi Function.

Basic idea.

Find all the elements which are relatively prime to both n and m , then it is relatively prime to mn . Formally:

$$\forall a, \gcd(a, n) = 1 \text{ and } \gcd(a, m) = 1 \implies \gcd(a, mn) = 1.$$

More about Euler's Phi Function.

Counting

- How many rows satisfy $\gcd(r, m) = 1$? Ans : $\varphi(m)$. Note that, if $\gcd(r, m) = 1$ then $\gcd(km + r, m) = 1$, for $k \in [0..n - 1]$.
- At r th row, how many integers have $\gcd(km + r, n) = 1$, for $k \in [0..n - 1]$? Ans: $\varphi(n)$. Note that, we have $\gcd(n, m) = 1$.
- Hence, there are $\varphi(m)$ rows, each containing $\varphi(n)$ integers relatively prime to mn .

More about Euler's Phi Function.

Remark (Why the conclusion holds?)

If $\gcd(a, m) = 1$ and $\gcd(a, n) = 1$ then $\gcd(a, mn) = 1$.

More about Euler's Phi Function.

Remark (Why the second item holds?)

The elements in r th row are: $r, m + r, \dots, (n - 1)m + r$ with $\gcd(m, n) = 1$. $\forall k_i \neq k_j, k_i m + r \not\equiv k_j m + r \pmod{n}$. Otherwise, $k_i = k_j$ by our Golden Law (Cancellation Law), contradiction! It means the n elements in r th row form "a complete system of residues modulo n ", that is $\{r, m + r, \dots, (n - 1)m + r\} \pmod{n} = \{0, 1, 2, \dots, n - 1\}$. Hence, exactly $\varphi(n)$ of these integers are relatively prime to n .

Theorem

Let m and n be relatively prime positive integers. Then $\varphi(mn) = \varphi(m)\varphi(n)$.

More about Euler's Phi Function.

Some easy generalizations.

How to relate $\varphi(mn) = \varphi(m)\varphi(n)$ where $\gcd(m, n) = 1$ with $\varphi(m)$ where $m = p^i q^j$?

More about Euler's Phi Function.

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Ans: $\gcd(p^i, q^j) = 1$ when p and q are relatively prime. Hence $\varphi(p^i q^j) = \varphi(p^i)\varphi(q^j)$.

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How to generalize the result to $\varphi(m)$ where $m = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$?

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Ans: Induction! Left as an exercise.

习题.

求 a , 使得 a 满足以下条件:

- $a \equiv 7^{2003} \pmod{3750}$
- $1 \leq a \leq 5000$
- a 不被 7 整除.

习题.

使用费尔马小定理求解同余方程 $x^{51} \equiv 2 \pmod{17}$ 和 $x^{50} \equiv 2 \pmod{17}$ 。

习题.

请求解 $x^{113} \equiv 2 \pmod{221}$ 。[提示: 113 与 $\varphi(221)$ 互素。]

Prove the following theorem.

Theorem (Euler's Phi Function.)

Let $m = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$, for all p_i ($1 \leq i \leq k$) is a prime, then $\varphi(m) = m(1 - 1/p_1)(1 - 1/p_2) \cdots (1 - 1/p_k)$.

- $pk, sk \leftarrow KG(\lambda)$, sk is local and secret, while pk is public.
- Given plaintext m , compute ciphertext $c \leftarrow Enc(m, pk)$
- Given ciphertext c , compute plaintext $m \leftarrow Dec(c, sk)$

- $pk, sk \leftarrow KG(\lambda)$:
 - Choose two large primes p and q , $n = p \cdot q$, let $\varphi(n) = (p-1)(q-1)$.
 - Choose a integer e , with $\gcd(e, \varphi(n)) = 1$, compute d , s.t. $ed \equiv 1 \pmod{\varphi(n)}$.
 - Let pk be e and n ; sk be d and n .
- $Enc(m, pk)$: let $c \equiv m^e \pmod{n}$
- $Dec(c, sk)$: let $m \equiv c^d \pmod{n}$

Why RSA is correct?

Because of Euler's Theorem.

- $KG(\lambda)$:
 - e and d satisfy $ed \equiv 1 \pmod{\varphi(n)}$.
 - means: $\exists k$, s.t. $ed = k\varphi(n) + 1$
- $Enc(m, pk)$: $c \equiv m^e \pmod{n}$
- $Dec(c, sk)$: $c^d \equiv m^{ed} \equiv m^{k\varphi(n)+1} \equiv m \pmod{n}$

- $KG(\lambda)$:
 - Choose two primes $p = 7$ and $q = 11$, $n = p \cdot q = 77$, let $\varphi(n) = (p - 1)(q - 1) = 60$.
 - Choose a integer $e = 7$, with $\gcd(e, \varphi(n)) = 1$, compute d , s.t. $ed \equiv 1 \pmod{\varphi(n)}$, $d = 43$.
 - Let pk be $e = 7$ and $n = 77$; sk be $d = 43$ and $n = 77$.
- $Enc(m, pk)$: Given $m = 3$, then $c = 3^7 \bmod 77 = 31$.
- $Dec(c, sk)$: Let $m = 31^{43} \bmod 77 = 3$.

What is a Concrete Introduction?

- Play with numbers.
- Find the patterns, find the fun.
- Programming is a good way to play.

What have been covered?

- Congruence.
- Fermat's little theorem.
- Euler's theorem.

What have been omitted?

- Fast multiplication.
- Powers: how to do fast power.

What is the next step?

- From a new perspective to view Fermat's and Euler's theorems.
- From arithmetic go to algebra.